SPR Optimizer Project Analysis

1. Understanding the Codebase

The project is a simulation and optimization tool for Surface Plasmon Resonance (SPR) biosensors, designed to maximize sensitivity using AI.

- **physics.py (The Engine):** Uses the Transfer Matrix Method (TMM) to calculate the Reflectance Curve and Figure of Merit (FOM). It uses Numba JIT for high-speed processing.
- **materials.py (The Database):** Stores complex refractive indices for the Prism (SF10), Metals (Ag, Au, Cu), and Dielectrics.
- **optimizer.py (The Brain):** Uses Optuna to perform Bayesian optimization. It minimizes -FOM to find the optimal thickness for layers D2, D3, and D4.
- **postprocess.py (The Analyst):** Extracts trial history from the SQLite database and generates CSV exports and visualization plots.

2. Understanding the Results

Analysis based on 'best_trials.csv' and generated plots:

- **The Best Result:** A FOM of ~187 was achieved using Gold (Au) with specific layer thicknesses. This indicates a highly sensitive sensor configuration.
- **Optimization History:** The optimizer converged to the global optimum very quickly (within the first ~50 trials), spending the rest of the time exploring without finding better results.
- **Parameter Importance:** Analysis shows that 'D3_nm' (Dielectric thickness) is the most critical factor (43% importance), outweighing the metal choice.